Band-1 Hadamard Order 6724 — Miyamoto (Double Paley-I H_1680 + C_1682 over GF(41^2)) + Paley-II (q=3361) IN SPK; Gap 2052 Still OPEN

H_840 Paley-I PASS; H_1680 Double PASS; C_1682 GF(41^2) PASS; H_6724 Miyamoto PASS; H_6724 Paley-II PASS → IN SPK; gap 2052 still OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-15

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list · Human+AI attack order · ZoraASI mission-control.
Parents. `2026-09-15_Band1_Hadamard_6676_Miyamoto`; `2026-09-15_Band1_Hadamard_6628_Miyamoto`.
Lab. `/workspace/experiments/band1/hadamard_6724_miyamoto/`.
Cutoff. 2026-09-16T02:12:16Z (~10:12PM ET).

Honesty stamp (non-negotiable).
This memo claims **finite constructions of orders 840, 1680, and 6724 only** (verified $\max|HH^\top - nI| = 0$), via (i) Miyamoto with Double Paley-I seed H_{1680} = Sylvester-double(H_{840} = Paley-I($q = 839$)) and symmetric Paley conference C_{1682} over $q = 1681 = 41^2 \equiv 1 \pmod 4$, and (ii) classical Paley-II $q = 3361$ (prime $\equiv 1 \pmod 4$) → H_{6724}. Classical SPK: Paley-I **FAIL**s ($q = 6723 = 3^4 \cdot 83$); Paley-II **PASS**es ($q = 3361$); Kronecker **FAIL**s (4×1681). Therefore order **6724 is IN SPK via Paley-II** (Miyamoto is an additional finite construction, not the SPK reason). Triage slip note: $q = n/2 - 1 = 3361$, not $3362 = 2 \cdot 41^2$. It **does not** claim the full Hadamard conjecture. It **does not** stamp Clay-closed or finished-ToE. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved / Hadamard-conjecture-solved claims. Standard constructive gap **2052** remains **OPEN** under SPK. `clay_closed = false`. Prior skip $q = 1637 \to n = 6548$ retained; intervening conference-q after 1669: 1673/1677/1685/1689 not prime; $1681 = 41^2$ next admissible. Prior SPK seeds **6628** / **1668** / **1656** / **1620** / **1608** / **6436** retained; Miyamoto-only bonuses **6676/6484/6404/6388/6196/...** excluded; **840/1680/6724** folded into ONBOX when verified.

Abstract

Finite constructions of orders **840**, **1680**, and **6724**.

- H_{840} = Paley-I ($q = 839$) — Double half
- H_{1680} = Sylvester-double(H_{840}) — Miyamoto seed
- C_{1682} = Paley conference over GF(41^2), $q = 1681$
- H_{6724} = Miyamoto(H_{1680}, C_{1682})
- H_{6724} = Paley-II ($q = 3361$) — classical SPK
- Paley-I for 6724: **SKIP**

SPK: 6724 IN SPK via Paley-II q=3361. Seeds 840 / 1680 / 6724 enter SPK. Prior SPK **6628** / **1668** / **1656** / **1620** / **1608** / **6436** retained.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

`python3 verify_hadamard_6724.py`

Results. overall_pass=true; units 98/98; wall 496.3896s; all max|HHˆT-nI|=0.

Cutoff: 2026-09-16T02:12:16Z (~10:12PM ET)

Arithmetic

- Target $n = 6724 = 4 \cdot 1681$; $1681 = 41^2$ prime-power $\equiv 1 \pmod 4$ → Miyamoto-eligible.
- Need H_{1680} and C_{1682} (Paley conference over GF(41^2)).
- Seed: Paley-I $q = 839$ (prime $\equiv 3 \pmod 4$) → H_{840} → Sylvester-double → H_{1680}.
- SPK: Paley-I $q = 6723$ **FAIL**; Paley-II $q = 3361$ **PASS**; Kronecker **FAIL** → **IN SPK via Paley-II**.
- Prior skip: $q = 1637 \rightarrow n = 6548$. Intervening: 1673/1677/1685/1689 not prime; 1681=41^2 next admissible.

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_840 Paley-I	Paley-I q=839	0	0.621
H_1680 Double	Sylvester-double(H_840)	0	2.1195
C_1682	Paley conf GF(41^2) q=1681	0	2.7343
H_6724 Miyamoto	Miyamoto(H_1680 Double, C_1682)	0	238.9037
H_6724 Paley-II	Paley-II q=3361	0	245.5803
H_6724 Paley-I	SKIP q=6723=3 · 83 not prime	—	—

SPK closure note

Seeds include prior SPK ships through **6676** lineage (**6628** Paley-II; **1668** Paley-I; **1656** Double Paley-I; **1620**; **1608**; **6436**; **NOT** 6676/6484/6404/6388/6196) + verified **840** / **1680** / **6724** (Paley-II); **exclude** Miyamoto-only bonuses. Known count 1392. Next constructive gap **2052** (OPEN).

Quantity	Value
6724 in SPK	**True** (Paley-II q=3361)
1680 in SPK	**True** (Double Paley-I)
840 in SPK	**True** (Paley-I q=839)
1668 / 6628 / 1656 / 1620 / 1608 / 6436 in SPK	**True** (prior)
Next gap	**2052** OPEN

Non-claims

- No full Hadamard conjecture / Clay-closed / finished-ToE
- No RH / P NP / YM / Frankl / NS-solved / AC-solved
- Gap 2052 remains OPEN (method triage / non-finding ONLY)
- GATE-04 LOCKED; clay_closed=false

Repro

Lab: `/workspace/experiments/band1/hadamard_6724_miyamoto/`
Artifacts: `H840_paleyI_pm1.npz`, `H1680_double_pm1.npz`, `C1682_pm1.npz`, `H6724_miyamoto_pm1.npz`, `H6724_paleyII_pm1.npz`, `results.json`, `literature_ledger.json`, `SUMMARY.txt`.